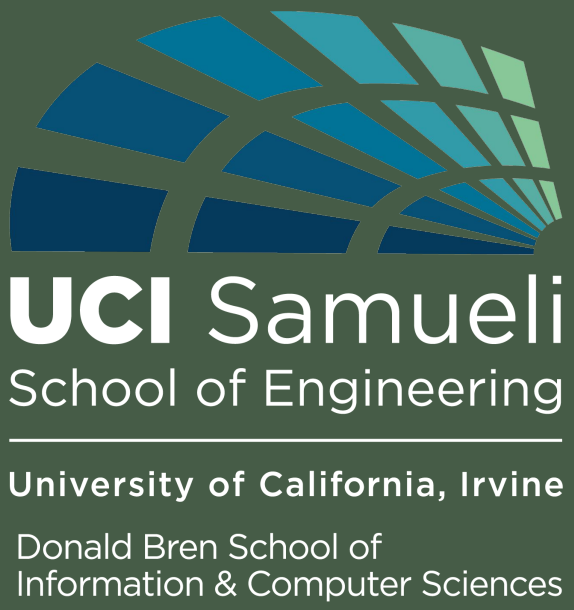




AI-Powered Free-Roaming Animatronic

Jaston Namahoe jpatao@uci.edu, Alex Garcia alexasg3@uci.edu, Ezequiel Villegas ezequv1@uci.edu, Bryan De La Luz Rojas delaluzb@uci.edu
Professor Nader Bagherzadeh
University of California, Irvine



ABSTRACT

Interactive animatronics in themed entertainment typically cost \$5,000–\$100,000+ [5], [6], limiting access for smaller parks and developers. While organizations such as Walt Disney Imagineering demonstrate highly immersive autonomous characters, these systems remain financially out of reach for most applications. This project presents an AI-powered free-roaming animatronic platform built from commercially available components, combining real-time speech recognition, contextual dialogue generation, and autonomous mobility. Survey results show 80% interest in interaction despite 80% reporting safety or reliability concerns, and the system achieves a 10×–200× cost reduction, demonstrating a scalable pathway for accessible interactive storytelling.

ADDITIONAL INFORMATION

- Hardware cost: ~\$500 (initial estimate: \$250; integration increased cost).
- A general interest survey was conducted with 10 participants to assess excitement and perceived engagement value.
- Operates indoors on flat terrain, not currently recommended for outdoor use, uneven surfaces, or weather conditions.
- Requires an intermediate level knowledge of robotics and programming.

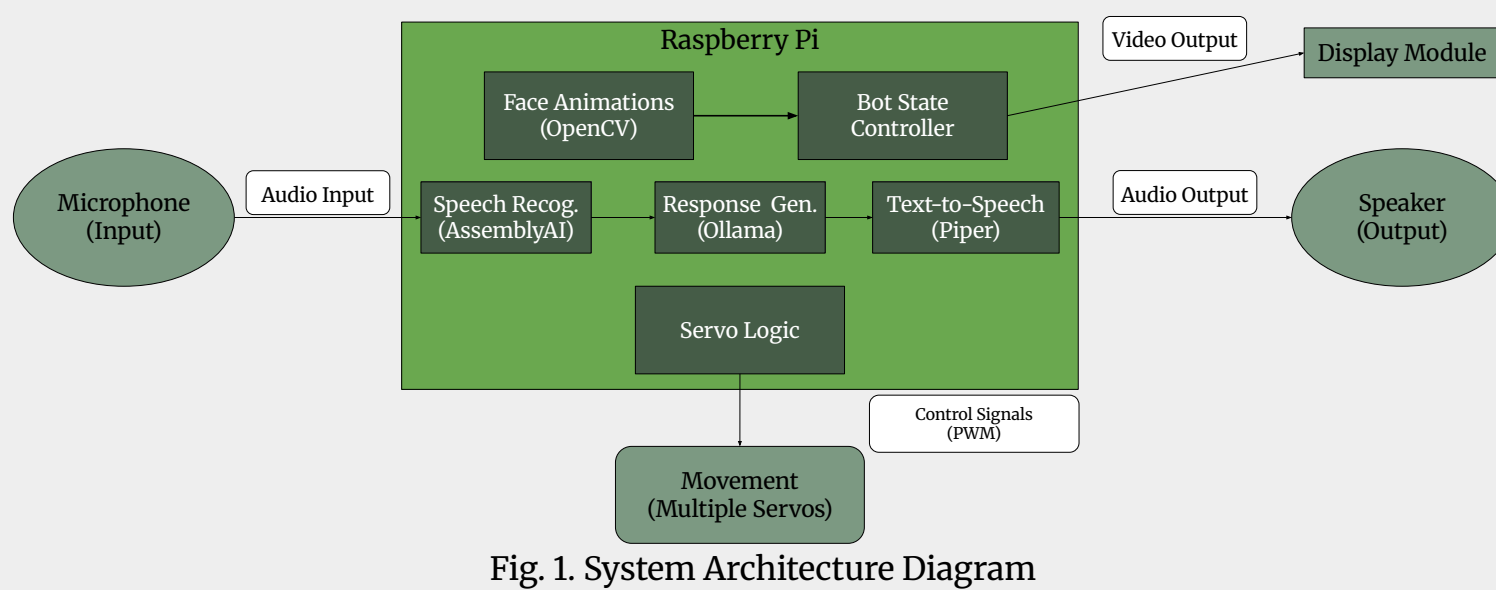


Fig. 1. System Architecture Diagram

OBJECTIVES

Building a free-roaming animatronic capable of safe, autonomous movement and delivering a versatile, adaptive character experience, this project has three main objectives:

- Utilize AI tools to generate real-time in-character responses, incorporating speech recognition for user detection and engagement, with a target of 75% of potential users expressing interest in interacting with the character.
- Access and respond to questions with relevant, situation-specific information.
- Design a mechanism for machine movement that is affordable and easy to reproduce with a total hardware cost of \$250–500, representing a substantial cost reduction to commercial animatronic systems.

METHODS

Real-Time Interaction

- Ollama (qwen3.5 for real-time responses): Local Large Language Model that performs natural language processing and generation [2].
- Piper (voice module for speech synthesis): Local text-to-speech engine that converts text into spoken audio with low latency, and provides support for multiple voices and languages [3].
- AssemblyAI (speech recognition for real-time interaction): Cloud-based automatic speech recognition service that converts live audio input into text for AI processing and response generation [4].

Context Awareness

- Queue Times API (Interface for live waiting time data): Real-time queue wait time and ride status API that enables dynamic context awareness based on current crowd conditions and ride wait data [7].

Mechanics

- Raspberry Pi 5 (main controller for hardware components): Single-board computer that runs the AI and coordinates servo motors, speakers, and display module.
- Fusion 360 (CAD software for mechanical design): Computer-aided design platform used to model, prototype, and refine the animatronics' structural and mechanical components, and movement mechanisms.

RESULTS

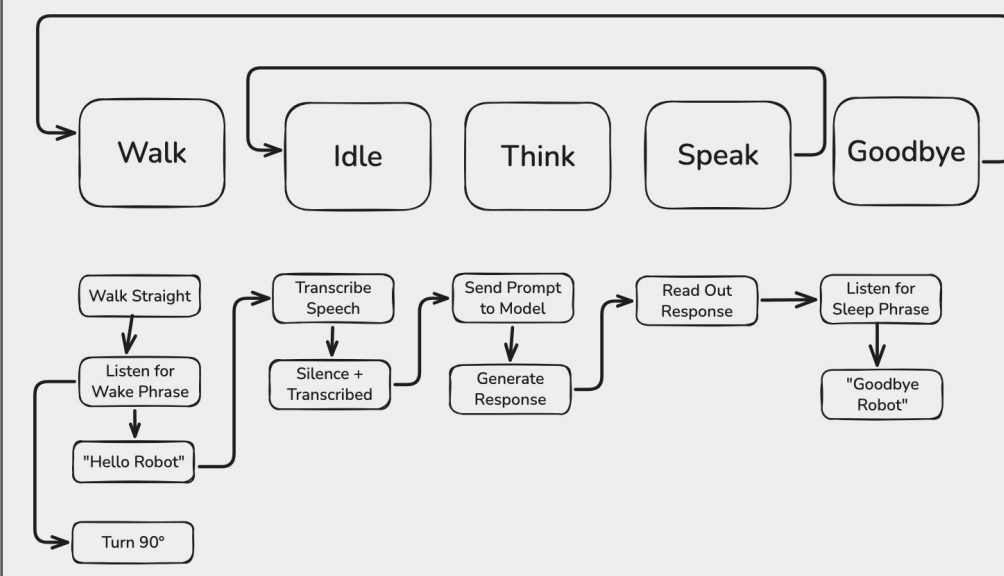


Fig. 2. Robot Operational States



Fig. 3. CAD Model of Leg Mechanism



Fig. 4. Fully Assembled Free-Roaming Animatronic Prototype

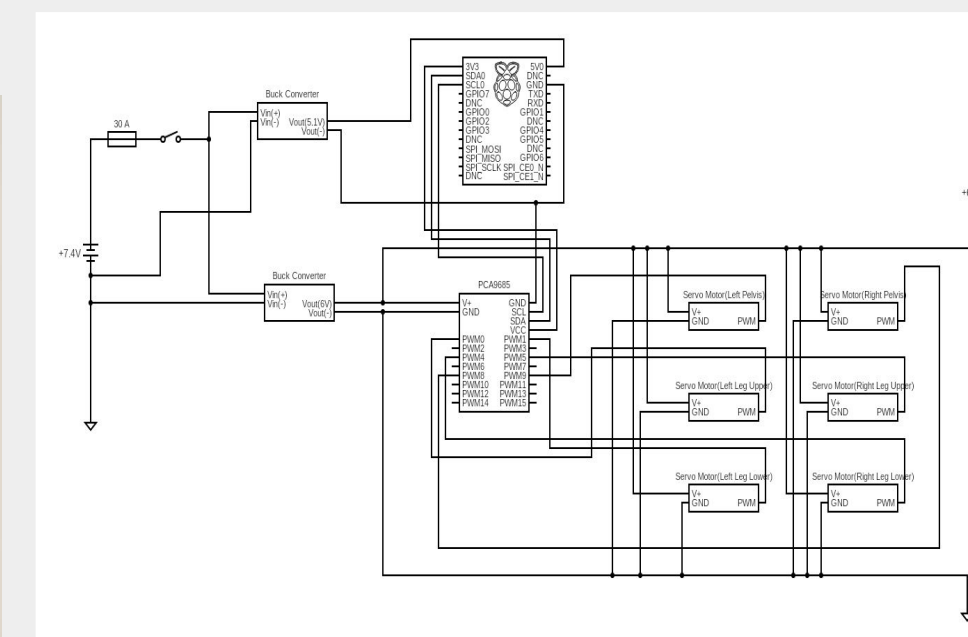


Fig. 5. System Circuit Diagram

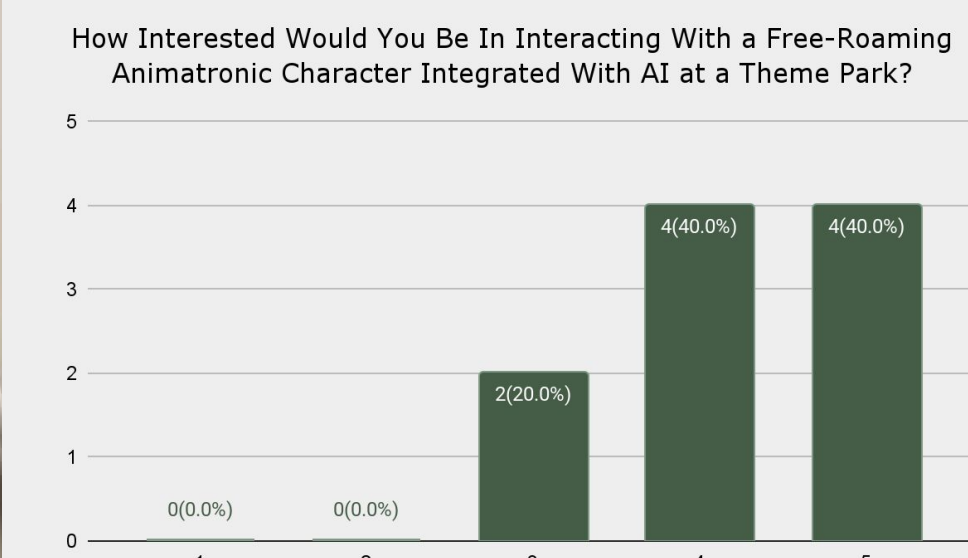


Fig. 6. Survey Results: Interaction Interest



Fig. 7. System Demonstration

CONCLUSIONS

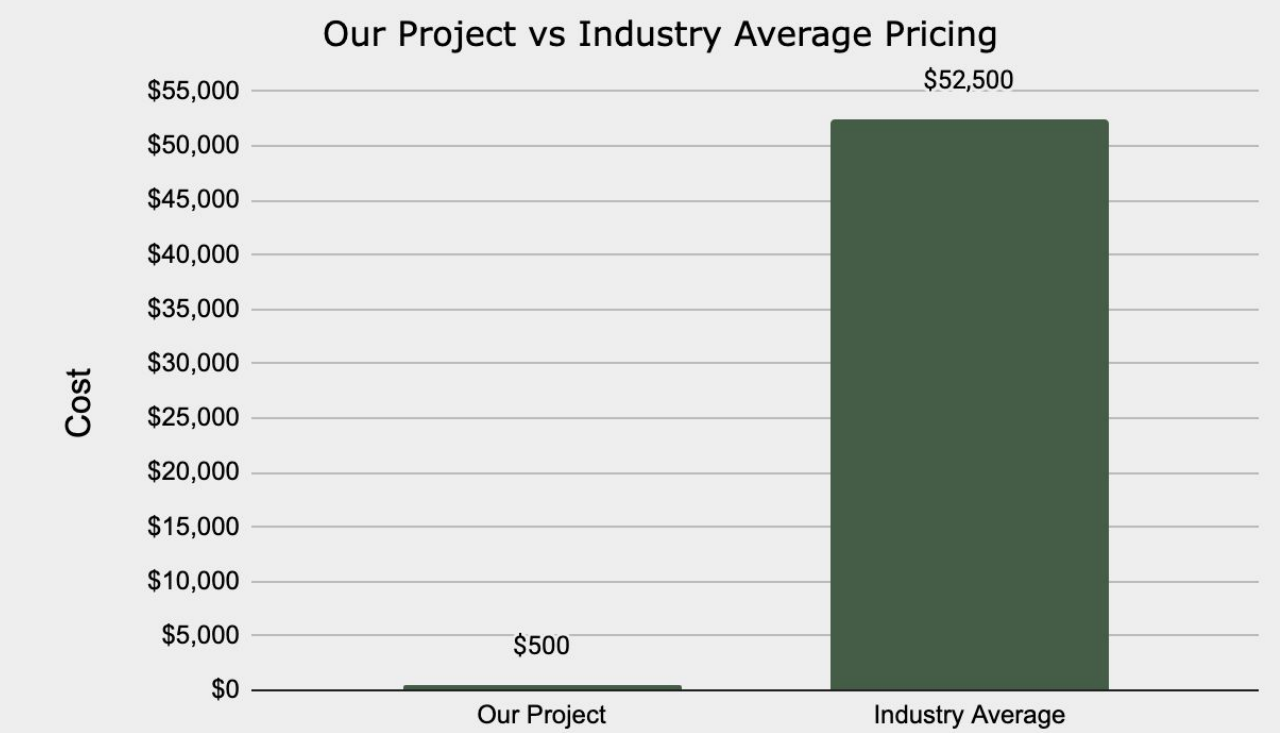


Fig. 8. Cost Comparison with Industry Standards

- Comparable technologies typically range from \$5,000–\$100,000 [5], [6].
- 10x–200x lower cost than comparable commercial systems.

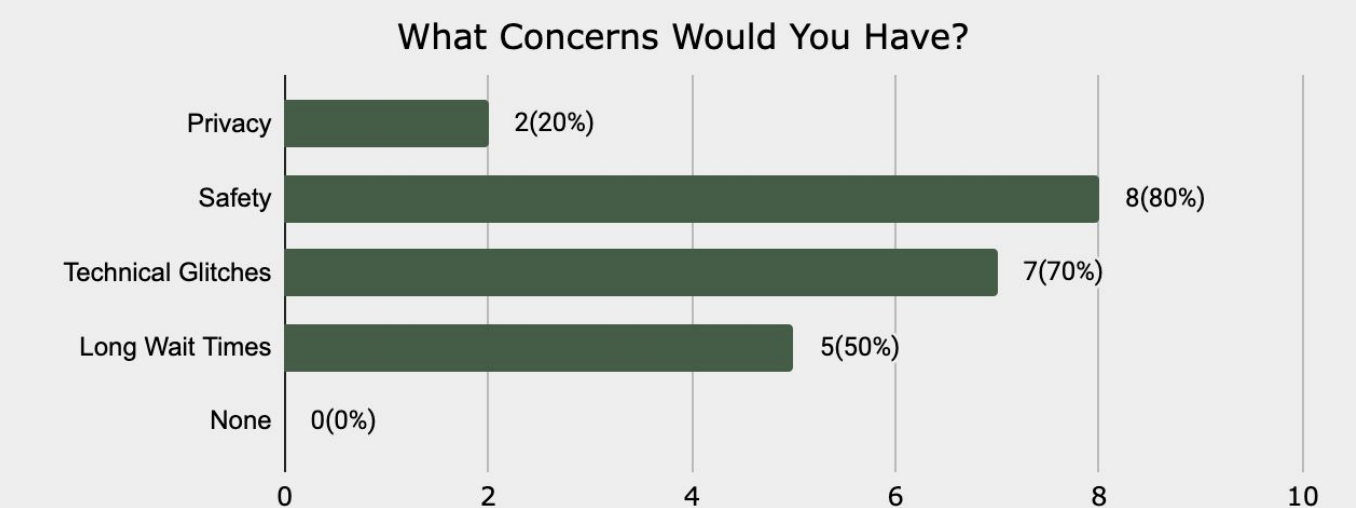


Fig. 9. Survey Results: Safety and Reliability Concerns

- 80% of respondents indicated a strong interest in interacting with an AI-powered, free-roaming animatronic.
- Indicates strong perceived novelty and engagement value.
- 80% expressed safety concerns (movement & collision risk).
- 20% concerned about data privacy.
- Indicates trust and safety are primary adoption barriers among users despite strong interest.

REFERENCES

- [1] J. Bruton, "Building a better Star Wars AT-AT toy," *Youtube*, Jun. 12, 2026. https://youtu.be/AAGPmHINRJA?si=55F5-7H3FmaDT_Zb (accessed Feb. 22, 2026).
- [2] Ollama, "Ollama's documentation - Ollama," *Ollama.com*, 2025. <https://docs.ollama.com/>
- [3] M. Hansen, "GitHub - OHF-Voice/piper1-gpl: Fast and local neural text-to-speech engine," *GitHub*, Mar. 28, 2025. <https://github.com/OHF-Voice/piper1-gpl?tab=readme-ov-file> (accessed Feb. 22, 2026).
- [4] Y. Tang, "Real-time speech recognition with Python," *AssemblyAI.com*, Sep. 17, 2025. <https://www.assemblyai.com/blog/real-time-speech-recognition-with-python> (accessed Feb. 22, 2026).
- [5] "FAQ," *Garnerholt.com*, 2025. <https://garnerholt.com/ghp-inc/about-us/faq.aspx>
- [6] "How Much Does an Animatronic Cost? A Comprehensive Guide for Enthusiasts in 2025 - CineMagic Workshop - Innovative Animatronics Manufacturer & Creative Solutions," *CineMagic Workshop - Innovative Animatronics Manufacturer & Creative Solutions*, May 10, 2025. <https://cinemagicworkshop.com/how-much-does-an-animatronic-cost-a-comprehensive-guide-for-enthusiasts-in-2025/> (accessed Feb. 24, 2026).
- [7] "Theme Park Wait Times API," *Queue-times.com*, 2021. <https://queue-times.com/en-US/pages/api> (accessed Feb. 25, 2026).